Forecasting the Equity Risk Premium Out-of-Sample: Regularized Machine Learning, Technical Indicators and Tactical Asset Allocation

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Abstract

We re-examine out-of-sample predictability of the US equity premium using a fixed 240-month rolling-window design on data spanning February 1951 to December 2025. The predictor set combines the fourteen Goyal–Welch (2008) macroeconomic variables (with one to six monthly lags) and the fourteen technical indicators of Neely, Rapach, Tu and Zhou (2014), which we reproduce from raw price and volume data with a 99.98% signal-month match to the published series. Against the prevailing historical mean benchmark we evaluate ℓ_1-penalized least squares (LASSO and post-LASSO), Ridge, Elastic Net, principal components and partial least squares regressions, and random forests, together with equal-weight and constrained Granger–Ramanathan forecast combinations, at horizons $h \in \{1, 3, 12\}$. Statistical accuracy is assessed with the Diebold–Mariano (1995) test on squared and absolute errors, and economic value with a binary switching strategy and a mean–variance investor, both net of transaction costs. Our expanding-window comparison uses 178 correctly timed predictor–target pairs before the January 1966 forecast, versus the 181-month initial window in the original study. Our computed technical signals match the published series in 99.98% of signal-months. On the 1966–2011 sample, the three focal volume-rule estimates are positive, but none of the six volume rules is significant at 5%. On the extended sample, however, no model beats the prevailing mean at any horizon: out-of-sample R^2 ranges from -15.49% to -0.38%, while hit rates are close to the benchmark. At $h = 1$, the LASSO switching strategy has lower terminal wealth and only modestly smaller maximum drawdown than buy-and-hold, while mean–variance utility differences are negative for all forecast models. At $h = 1$ and 10 bps net, ΔU is -0.39% p.a. for LASSO and -2.07% for post-LASSO. The corrected results provide no evidence that these forecasts improve on the prevailing mean in point-forecast accuracy or mean–variance utility.

1 Introduction

Whether the equity premium is predictable out-of-sample is one of the most debated questions in empirical finance. Goyal and Welch (2008) showed that a broad set of macroeconomic variables fails to beat the historical average in real time, and Goyal, Welch and Zafirov (2024) reach the same verdict with a greatly expanded predictor set. Against this backdrop, Neely, Rapach, Tu and Zhou (2014) report that *technical indicators*—moving average, momentum and on-balance volume rules—do contain out-of-sample predictive content for the premium on their 1966–2011 evaluation period, and Gu, Kelly and Xiu (2020) document the promise of machine learning methods in asset pricing more generally.

This paper asks a simple question: once the technical-indicator information of Neely et al. (2014) is combined with the Goyal–Welch macro variables and fed through modern regularized machine learning models, does anything beat the prevailing historical mean on the extended 1951–2025 sample? We follow a disciplined protocol: a fixed 240-month rolling window, a strict no-lookahead alignment convention, the prevailing mean as the only benchmark, Diebold–Mariano (1995) tests with HAC inference, and economic evaluation net of transaction costs. To validate the data, we match the published technical signals in 99.98% of signal-months, then use a time-aligned expanding-window bivariate design on the 1966–2011 sample. Its correctly timed predictor– target pairs provide 178 initial observations, compared with 181 months in the original design.

Three findings emerge. First, on the extended sample no model—LASSO, post-LASSO, Ridge, Elastic Net, PCR, PLS, random forest, or forecast combinations—beats the prevailing mean at any horizon; out-of-sample R^2 ranges from -15.49% to -0.38%. Second, the three focal volume-rule estimates are positive but do not reach 5% significance on the 1966–2011 sample, and none of the six volume rules is significant at that level; the three focal estimates are negative on the extension. Third, at $h = 1$ the LASSO switching strategy has lower terminal wealth than buy-and-hold and a modestly smaller maximum drawdown, while forecast-based mean–variance utility differences are negative. At $h = 1$ and 10 bps net, ΔU is -0.39% p.a. for LASSO and -2.07% for post-LASSO. The remainder of the paper describes the data (Section 2), the methods (Section 3), the results (Section 4) and concludes in Section 5.

2 Data

2.1 Sources and sample

The target variable is the monthly log equity premium on the S&P 500, $y_t = 100\,(\ln(1 + R_t^m) - \ln(1 + R_t^f))$, where R_t^m is the value-weighted S&P 500 return including dividends and R_t^f the risk-free rate, taken from the continuously updated Goyal–Welch–Zafirov (2024) dataset (monthly observations 1871:01–2025:12). The premium is expressed in percent per month; over the estimation sample its mean is 0.66% with a standard deviation of 4.21%. The fourteen macroeconomic predictors follow Goyal and Welch (2008): dividend-price ratio, dividend yield, earnings-price ratio, dividend payout ratio, stock variance, book-to-market, net equity expansion, Treasury bill rate, long-term yield, long-term return, term spread, default yield spread, default return spread, and inflation. Each macro predictor enters with one to six monthly lags.

The fourteen technical indicators follow Neely et al. (2014): six moving average rules $\mathrm{MA}(j, m)$ with $j \in \{1, 2, 3\}$ and $m \in \{9, 12\}$ comparing short and long moving averages of the index level, two momentum rules $\mathrm{MOM}(m)$, $m \in \{9, 12\}$, and six on-balance volume rules $\mathrm{VOL}(j, m)$ comparing moving averages of on-balance volume. Each indicator is a binary signal (1 = hold equities). We compute the signals from the raw index level and monthly S&P 500 volume: the paper's own volume series for 1950–2011 and monthly average daily volume from Yahoo Finance for 2012–2025 (the untruncated Yahoo series correlates 0.9999 with the paper series on their 1985–2011 overlap, and the volume-based signals are invariant to a constant volume rescaling). Against the published signals of Neely et al. (2014) over their 1950:12–2011:12

sample, 12 of the 14 signals match exactly and the remaining two match in 99.9% of months; the two mismatching months are documented data revisions.

2.2 Alignment and sample period

Row t of the panel holds the premium *earned* in month t together with predictors *observable at the end of month* t. Macro lag columns satisfy $x_t^{\text{lag}\,k} = x_{t-k}$, $k = 1, \ldots, 6$; technical signals at row t use price and volume data through month t only, so no forecast uses information unavailable at its origin. The sample is the first month in which every column is non-missing (1951:02, driven by the twelve-month burn-in of the volume rules) through 2025:12, giving $T = 899$ monthly observations. Table 1 reports summary statistics and augmented Dickey–Fuller (BIC lag selection, intercept and trend) tests.

3 Methodology

3.1 Rolling-window forecasting design

All models are estimated on a fixed 240-month (20-year) rolling window and produce an h-month-ahead forecast of the premium, $h \in \{1, 3, 12\}$. A training observation pairs predictors X_s, observable at the end of month s, with the premium earned in month $s + h$. At forecast origin t, training pairs are restricted to outcomes observed by t; the window contains 240 valid pairs. The benchmark is the prevailing (historical) mean premium computed through the forecast origin, the only admissible benchmark for this exercise: OLS benchmark forecasts fail out-of-sample and are not used.

3.2 Models

The predictor matrix is wide relative to the 240-month estimation window, its columns are highly correlated (six lags of fourteen persistent macro series plus fourteen signals that switch in clusters), and any true signal is likely sparse and episodic. This is precisely the setting for which the penalized and dimension-reduction estimators below are designed: the ℓ_1/ℓ_2 penalties tame the collinearity, the component methods condense the factor structure, and the random forest admits nonlinear interactions and threshold effects that linear models cannot represent. We estimate seven models within each window: (i) LASSO with the theoretical, data-driven penalty of `hdm::rlasso` (no cross-validation required); (ii) post-LASSO, i.e. OLS refit on the LASSO-selected support; (iii) Ridge and (iv) Elastic Net ($\alpha = 0.5$) with the penalty chosen by ten-fold cross-validation *inside the training window* (no information from the test month is ever used); (v) principal components regression and (vi) partial least squares with the number of components selected by cross-validated PRESS (scaled inputs, at most 20 components); and (vii) a random forest (500 trees, $m_{\text{try}} = p/3$, minimum node size 5, fixed seed, single-threaded for exact reproducibility). All models use the same 98-predictor matrix (84 macro-lag columns plus 14 technical signals); the expanded Goyal–Welch–Zafirov (2024) predictors are held out as a robustness variant. That variant adds only the eight expanded variables that are observed monthly throughout 1951:02–2025:12 and are not constructed with full-sample

Table 1: Summary statistics and unit-root tests (sample: 1951:02–2025:12).

Series	Mean	SD	Min	Max	Skew	Kurt	ADF	Sig.
Panel A: simple equity premium (%, monthly)								
y_t (full sample)	0.662	4.211	-22.18	16.30	-0.42	4.54	-29.18	1%
Panel B: macroeconomic predictors (raw series)								
d/p	0.030	0.013	0.01	0.07	0.69	2.91	-2.82	>10%
d/y	0.030	0.013	0.01	0.07	0.69	2.91	-2.70	>10%
e/p	0.063	0.026	0.01	0.15	0.98	3.58	-3.57	5%
d/e	0.500	0.283	0.29	3.97	8.48	88.54	-5.92	1%
svar	0.002	0.005	0.00	0.07	10.87	149.96	-19.89	1%
b/m	0.486	0.250	0.12	1.21	0.70	2.72	-1.99	>10%
ntis	0.011	0.020	-0.06	0.05	-0.62	2.88	-4.19	1%
tbl	0.041	0.030	0.00	0.16	0.92	4.25	-2.87	>10%
lty	0.057	0.028	0.01	0.15	0.83	3.30	-1.83	>10%
ltr	0.005	0.027	-0.11	0.15	0.52	6.05	-28.18	1%
tms	0.016	0.014	-0.04	0.05	-0.04	2.75	-4.74	1%
dfy	0.009	0.004	0.00	0.03	1.89	8.25	-4.39	1%
dfr	0.000	0.014	-0.10	0.07	-0.66	10.97	-23.79	1%
infl	0.003	0.004	-0.02	0.02	0.14	5.54	-3.19	10%
Panel C: technical signals (binary; mean = fraction of months in "buy")								
MA_1_9	0.709							
MA_1_12	0.729							
MA_2_9	0.710							
MA_2_12	0.730							
MA_3_9	0.715							
MA_3_12	0.731							
MOM_9	0.729							
MOM_12	0.743							
VOL_1_9	0.694							
VOL_1_12	0.720							
VOL_2_9	0.688							
VOL_2_12	0.719							
VOL_3_9	0.704							
VOL_3_12	0.718							

ADF: intercept and trend, common-sample BIC lag selection; MacKinnon (2010) asymptotic critical values $-3.959/-3.410/-3.127$ (1%/5%/10%). $T = 899$ monthly observations before ADF lags.
Skewness and kurtosis are population moments; kurtosis is NOT excess (a normal distribution has kurtosis 3).
The premium is $100\,(R^m - R^f)$ using monthly simple returns; macro predictors enter the models with one to six lags; technical signals are the 14 binary rules of Neely et al. (2014).

information (skvw, tail, rdsp, avgcor, dtoy, dtoat, wtexas, lzrt), each with six lags, giving 146 predictors. We exclude fbm, ogap, sntm, tchi and shtint because the data documentation describes them as full-sample constructions, so lagging them would not remove look-ahead bias; twelve further variables (e.g. cay, pce, eqis) are not available monthly; and csp, vp, impvar, vrp, ndrbl, ygap, rsvix and disag start after or end before our sample, so including them would shorten the common sample. The expanded variant is therefore a partial, not exhaustive, use of the Goyal et al. (2024) predictor set.

3.3 Forecast combinations

We combine the seven model forecasts with (i) the simple equal-weight average and (ii) a constrained Granger–Ramanathan (1984) regression: at each origin the weights solve $\min_w \|y - Fw\|^2$ subject to $\sum_i w_i = 1$ and $w_i \geq 0$, estimated only on past out-of-sample forecast errors known at that origin (targets $\leq$ origin). Before 60 usable observations accumulate, equal weights are used.

3.4 Evaluation

Statistical accuracy is measured by out-of-sample RMSE and MAE and the Campbell–Thompson (2008) $R^2_{\text{OOS}} = 1 - \sum e_t^2 / \sum e_{t,\text{PM}}^2$ against the prevailing mean. Significance is tested with the Diebold–Mariano (1995) statistic on squared and absolute error differentials, with Newey–West long-run variance (lag $h - 1$). Following the course, we use the Diebold–Mariano test only and do not report the Clark–West (2007) nested-model test. Economic value is assessed with two strategies net of transaction costs (baseline 10 bps, sensitivity 5 and 20 bps). Each switch, including the initial equity purchase, is charged this cost. We assume the investor trades an S&P 500 ETF and Treasury bills; major US brokers now charge zero online commission on both (e.g. Charles Schwab, 2026), and a one-cent quoted spread on a liquid S&P 500 ETF priced in the hundreds of dollars is well below 1 bp. The 10 bps baseline is therefore a conservative all-in cost covering spread, slippage and price impact, and the 20 bps case reflects the higher trading costs of earlier decades: (i) a binary switching strategy that holds the S&P 500 when the h-month-ahead premium forecast is positive and three-month Treasury bills otherwise, reporting terminal wealth, annualized Sharpe ratio, maximum drawdown, and turnover; and (ii) a mean–variance investor with risk aversion $\gamma = 3$ who allocates $w_t = (1/\gamma)\,\hat{y}_t/\hat{\sigma}_t^2$ to equities, clamped to $[0, 1.5]$, where $\hat{\sigma}_t^2$ is realized variance over the previous 60 months, measured using information available at the forecast origin; the certainty-equivalent return gain ΔU relative to the benchmark allocation is reported.

4 Results

4.1 Statistical accuracy

Table 2 reports out-of-sample RMSE, MAE, R^2_{OOS}, hit rate, and the Diebold–Mariano p-values of every model against the prevailing mean at each horizon, and Figure 1 visualizes the R^2_{OOS} profile. The central result is unambiguous: *no model beats the prevailing mean.* All nine forecast

models have negative out-of-sample R^2 at each horizon. At $h = 1$, the prevailing-mean RMSE is 4.403; LASSO has the least negative R^2_{OOS} (-1.06%), while random forest has the most negative (-15.49%). Post-LASSO is closest to the benchmark at $h = 3$ (-0.38%) and $h = 12$ (-0.52%). Across all horizons, the one-sided Diebold–Mariano tests find no statistically significant evidence that a model improves on the benchmark (minimum one-sided $p = 0.677$). This does not support the previous claim that models are significantly worse. Hit rates range from 53.5% to 60.1%, compared with prevailing-mean rates of 59.7% to 59.9%; thus, the direction metric is close to the benchmark and is not uniformly lower. The full, technical-only and macro-only predictor-set results are negative as well (Appendix A).

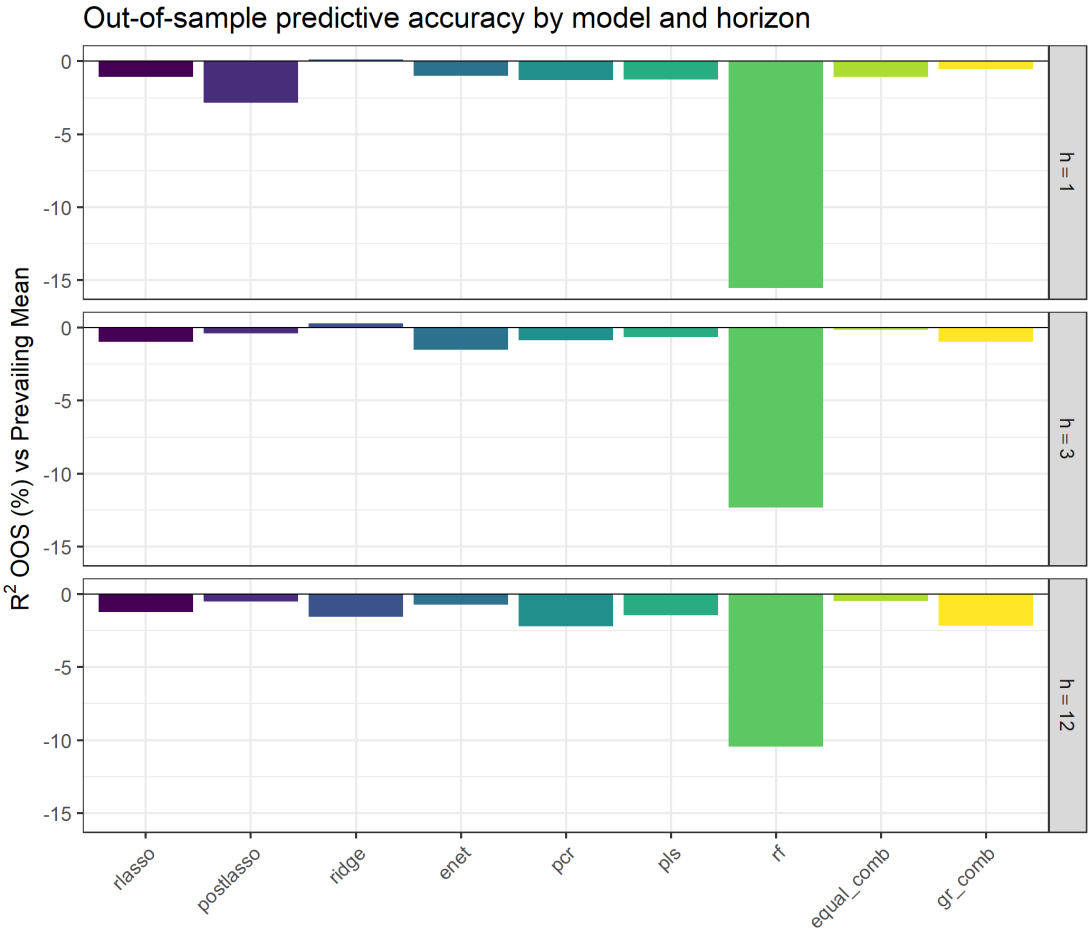


Figure 1: Out-of-sample R^2 (%) versus the prevailing mean by model and horizon. Negative values indicate lower accuracy than the benchmark.

4.2 Replicating Neely et al. (2014)

To examine the individual technical signals, we estimate the expanding-window bivariate predictive regressions of the one-month-ahead premium on one technical signal at a time—on their 1966–2011 evaluation sample and on the 2012–2025 extension (Table 3). With predictor signals dated before the return they forecast, the three focal on-balance-volume estimates are VOL(2,9) +0.19% (one-sided Diebold–Mariano $p = 0.339$), VOL(2,12) +0.11% ($p = 0.403$), and VOL(3,12) +0.53% ($p = 0.196$). None of these three focal estimates is significant at 5%;

Table 2: Out-of-sample accuracy vs. the prevailing mean benchmark.

Model	RMSE	MAE	R^2_{OOS} (%)	HR	DM_{RMSE}	p_{DM}	p_{MAE}
Panel: $h = 1$ (OOS $n = 658$; PM hit rate 0.60)							
rlasso	4.426	3.344	-1.06	0.59	1.05	0.852	0.716
postlasso	4.465	3.348	-2.83	0.60	1.76	0.961	0.755
ridge	4.400	3.323	0.12	0.60	-0.12	0.451	0.233
enet	4.425	3.343	-1.00	0.60	1.39	0.917	0.745
pcr	4.431	3.336	-1.29	0.60	0.82	0.794	0.515
pls	4.430	3.344	-1.24	0.61	0.85	0.804	0.659
rf	4.732	3.555	-15.53	0.55	3.54	1.000	1.000
equal_comb	4.427	3.338	-1.09	0.59	0.80	0.788	0.550
gr_comb	4.414	3.326	-0.54	0.60	0.43	0.667	0.322
Panel: $h = 3$ (OOS $n = 654$; PM hit rate 0.60)							
rlasso	4.431	3.347	-0.97	0.58	1.18	0.880	0.683
postlasso	4.418	3.335	-0.38	0.60	0.46	0.677	0.415
ridge	4.403	3.329	0.30	0.60	-0.41	0.341	0.296
enet	4.442	3.366	-1.50	0.59	2.75	0.997	0.976
pcr	4.428	3.363	-0.88	0.57	0.64	0.740	0.819
pls	4.423	3.333	-0.64	0.61	0.48	0.684	0.394
rf	4.673	3.506	-12.33	0.54	2.49	0.994	0.991
equal_comb	4.412	3.338	-0.14	0.60	0.14	0.556	0.478
gr_comb	4.431	3.346	-0.97	0.59	0.87	0.809	0.624
Panel: $h = 12$ (OOS $n = 636$; PM hit rate 0.60)							
rlasso	4.470	3.391	-1.25	0.59	1.70	0.956	0.936
postlasso	4.454	3.382	-0.52	0.60	1.61	0.947	0.897
ridge	4.477	3.407	-1.54	0.59	1.98	0.976	0.984
enet	4.459	3.387	-0.72	0.59	1.29	0.902	0.893
pcr	4.491	3.404	-2.20	0.61	1.93	0.973	0.924
pls	4.474	3.394	-1.44	0.60	1.41	0.920	0.882
rf	4.669	3.560	-10.47	0.54	3.04	0.999	0.995
equal_comb	4.453	3.386	-0.48	0.58	0.89	0.814	0.853
gr_comb	4.490	3.417	-2.16	0.59	1.78	0.963	0.942

R^2_{OOS}: Campbell–Thompson (2008), in percent vs. the prevailing mean; negative values mean the model is less accurate than the benchmark. HR: hit rate, the fraction of months in which the forecast sign matches the realized sign (direction-of-change loss).

DM: Diebold–Mariano on squared errors, one-sided H1 "model more accurate"; p_{MAE}: absolute-error DM. Bartlett HAC bandwidth is $h - 1$, a maintained inference choice. $^{*}/^{**}/^{***}$: 10%/5%/1%.

none of the six volume rules is significant at that level either. On the 2012–2025 extension, the three focal estimates are negative and none is significant.

Table 3: Neely et al. (2014) replication and extension: expanding-window bivariate predictive regressions, one technical signal at a time, against the prevailing mean.

Sample	Signal	n	R^2_{OOS} (%)	DM p-value
Panel A: 1966–2011 (Neely et al. sample)				
	VOL(2,9)	552	0.19	0.339
	VOL(2,12)	552	0.11	0.403
	VOL(3,12)	552	0.53	0.196
	MA(2,12)	552	0.57	0.248
Panel B: 2012–2025 (extension)				
	VOL(2,9)	168	-1.51	0.952
	VOL(2,12)	168	-0.95	0.835
	VOL(3,12)	168	-0.31	0.603
	MA(2,12)	168	-1.09	0.776

Expanding estimation window with 178 correctly timed predictor–target pairs before the January 1966 forecast; one-sided Diebold–Mariano p-value (squared-error loss) against the prevailing mean. Complete results for all 14 signals are in `output/tables/table5_neely_replication.csv`; the signal-series comparison with the published series is in `output/tables/signal_validation.csv`. Estimates are our own time-aligned regressions, not values taken from the published study.

4.3 Economic evaluation

Table 4 reports the two economic strategies net of transaction costs, and Figure 2 plots the switching-strategy wealth paths. At $h = 1$, the LASSO-driven switching strategy ends with \$232.7 per dollar invested versus \$341.9 for buy-and-hold; its maximum drawdown is -44.4% versus -50.2%, and its annualized Sharpe ratio is 0.767 versus 0.776. These sample outcomes show lower terminal wealth and only a modestly smaller drawdown for this specification. The mean–variance utility differences relative to the prevailing-mean allocation are negative for every forecast model at $h = 1$ and at the reported fee levels. For example, at $h = 1$ and 10 bps net of transaction costs, ΔU is -0.39% p.a. for LASSO, -2.07% for post-LASSO, and -0.99% for the Granger–Ramanathan combination. The results therefore do not support a claim of positive mean–variance utility gains.

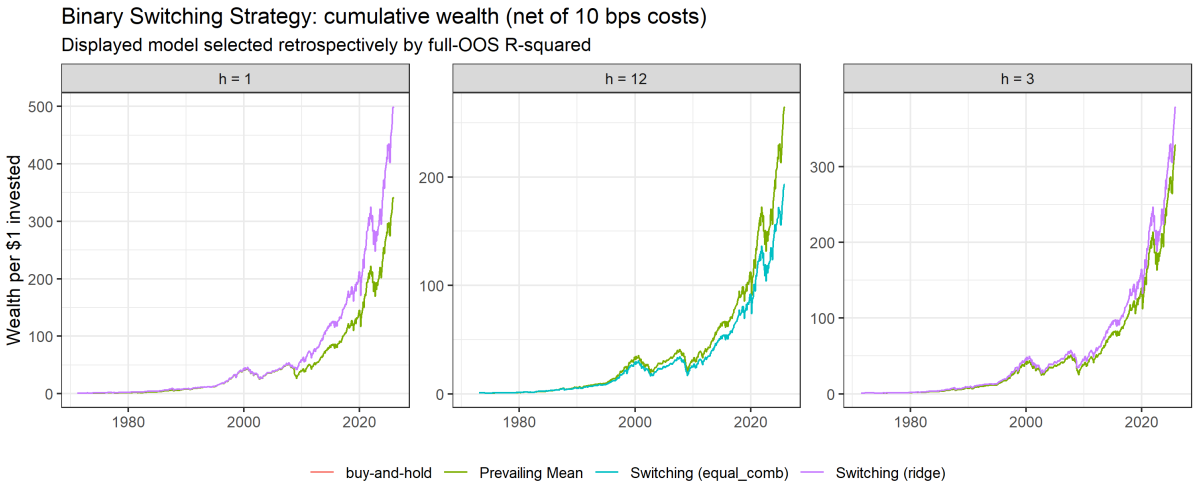


Figure 2: Cumulative wealth per \$1 of the binary switching strategy driven by the best model at each horizon (by R^2_{OOS}), the prevailing mean, and buy-and-hold, net of 10 bps transaction costs.

Table 4: Economic strategy outcomes net of transaction costs.

	Binary switching (10 bps)						Switching wealth		MV ΔU (% p.a.)			MV (10 bps)		
Model	Wealth	Buy-hold	Sharpe	MaxDD	Switch	Exp.	5 bps	20 bps	10 bps	5 bps	20 bps	CER	Sharpe	Mean w
Panel: $h = 1$														
rlasso	257.78	341.59	0.48	-0.44	35	0.95	262.33	248.90	0.08	0.18	-0.12	7.24	0.42	0.97
postlasso	242.67	341.59	0.46	-0.50	15	0.99	244.50	239.05	-0.86	-0.83	-0.92	6.30	0.36	0.91
ridge	499.36	341.59	0.57	-0.44	35	0.93	508.12	482.26	1.92	1.98	1.81	9.08	0.54	0.95
enet	345.52	341.59	0.51	-0.50	9	0.98	347.05	342.48	-0.23	-0.19	-0.32	6.93	0.40	0.93
pcr	330.38	341.59	0.51	-0.47	35	0.91	336.18	319.05	0.74	0.81	0.59	7.90	0.46	0.94
pls	398.41	341.59	0.54	-0.50	15	0.95	401.37	392.54	0.36	0.40	0.28	7.52	0.44	0.94
rf	92.58	341.59	0.41	-0.23	167	0.68	100.60	78.39	-1.96	-1.70	-2.47	5.21	0.27	0.79
equal_comb	382.92	341.59	0.54	-0.44	41	0.92	390.77	367.67	0.69	0.81	0.46	7.85	0.46	0.91
gr_comb	421.30	341.59	0.55	-0.44	35	0.92	428.73	406.82	1.18	1.25	1.06	8.35	0.49	0.96
Buy-and-Hold	341.59	341.59	0.49	-0.50	1	1.00								
Prevailing Mean	341.59	341.59	0.49	-0.50	1	1.00			0.00	0.00	0.00	7.16	0.42	1.07
Panel: $h = 3$														
rlasso	214.04	328.43	0.45	-0.45	25	0.96	216.71	208.78	-0.54	-0.46	-0.70	6.59	0.38	0.99
postlasso	336.86	328.43	0.50	-0.50	9	0.99	338.38	333.84	-0.35	-0.33	-0.41	6.77	0.39	0.92
ridge	378.79	328.43	0.52	-0.44	31	0.95	384.68	367.26	1.27	1.33	1.16	8.40	0.49	0.97
enet	255.73	328.43	0.47	-0.50	25	0.97	258.90	249.49	-1.80	-1.74	-1.93	5.33	0.29	0.92
pcr	220.28	328.43	0.47	-0.45	31	0.90	223.67	213.64	0.90	0.96	0.79	8.03	0.47	0.94
pls	505.54	328.43	0.57	-0.50	19	0.92	510.31	496.11	0.70	0.73	0.64	7.83	0.46	0.87
rf	212.56	328.43	0.54	-0.30	133	0.64	227.10	186.18	2.21	2.42	1.79	9.34	0.56	0.76
equal_comb	318.80	328.43	0.52	-0.44	45	0.91	326.02	304.81	0.96	1.06	0.75	8.08	0.48	0.89
gr_comb	304.91	328.43	0.51	-0.44	49	0.90	312.43	290.41	0.78	0.87	0.59	7.91	0.46	0.94
Buy-and-Hold	328.43	328.43	0.49	-0.50	1	1.00								
Prevailing Mean	328.43	328.43	0.49	-0.50	1	1.00			0.00	0.00	0.00	7.13	0.42	1.07
Panel: $h = 12$														
rlasso	148.54	264.75	0.42	-0.51	25	0.96	150.41	144.88	-0.89	-0.81	-1.05	5.72	0.34	0.93
postlasso	264.75	264.75	0.48	-0.50	1	1.00	264.89	264.49	-0.89	-0.88	-0.90	5.73	0.34	0.93
ridge	261.26	264.75	0.48	-0.50	7	0.98	262.16	259.47	-1.05	-1.02	-1.10	5.57	0.33	0.92
enet	264.29	264.75	0.49	-0.50	31	0.96	268.37	256.30	-0.42	-0.38	-0.51	6.19	0.36	0.93
pcr	323.26	264.75	0.52	-0.50	19	0.95	326.30	317.25	-0.26	-0.22	-0.35	6.35	0.37	0.93
pls	258.89	264.75	0.48	-0.50	9	0.99	260.04	256.62	-0.10	-0.07	-0.15	6.51	0.38	0.92
rf	76.28	264.75	0.38	-0.48	136	0.63	81.61	66.64	0.56	0.77	0.13	7.17	0.41	0.71
equal_comb	193.27	264.75	0.45	-0.50	39	0.95	197.05	185.92	-0.48	-0.38	-0.67	6.13	0.35	0.86
gr_comb	167.12	264.75	0.43	-0.50	45	0.91	170.91	159.80	-0.81	-0.70	-1.03	5.80	0.34	0.88
Buy-and-Hold	264.75	264.75	0.48	-0.50	1	1.00								
Prevailing Mean	264.75	264.75	0.48	-0.50	1	1.00			0.00	0.00	0.00	6.61	0.40	1.06

Switching: holds equity when the forecast is positive, monthly risk-free cash otherwise. Every strategy starts in cash, pays equity entry and subsequent turnover costs, and has no terminal liquidation. Switch counts include entry. MaxDD = maximum drawdown; Exp. = mean equity exposure; cash uses realized Rfree, not the annual tbl predictor. Each h-step forecast predicts one month's return h months ahead, not a cumulative return.
MV: mean–variance investor with $\gamma = 3$, $w_t \in [0, 1.5]$; ΔU = annualized certainty-equivalent gain over the benchmark allocation in percentage points; CER = certainty-equivalent return of the strategy (% p.a.).

4.4 Interpreting the sample comparisons

Subperiod and predictor-set comparisons are descriptive and do not identify why forecast performance differs. In particular, they cannot establish that crisis months caused predictability, or that passive flows and faster arbitrage explain any change over time. The corrected bivariate estimates do not reproduce 5% significance for the three focal volume signals on the 1966–2011 sample; the extension therefore does not establish that previously reported predictability subsequently decayed. The macro-only and technical-only comparisons in Appendix A report out-of-sample performance for those predictor sets; they do not establish causal mechanisms. Figure 3 and Table 5 report the underlying descriptive h=1 LASSO subperiod estimates.

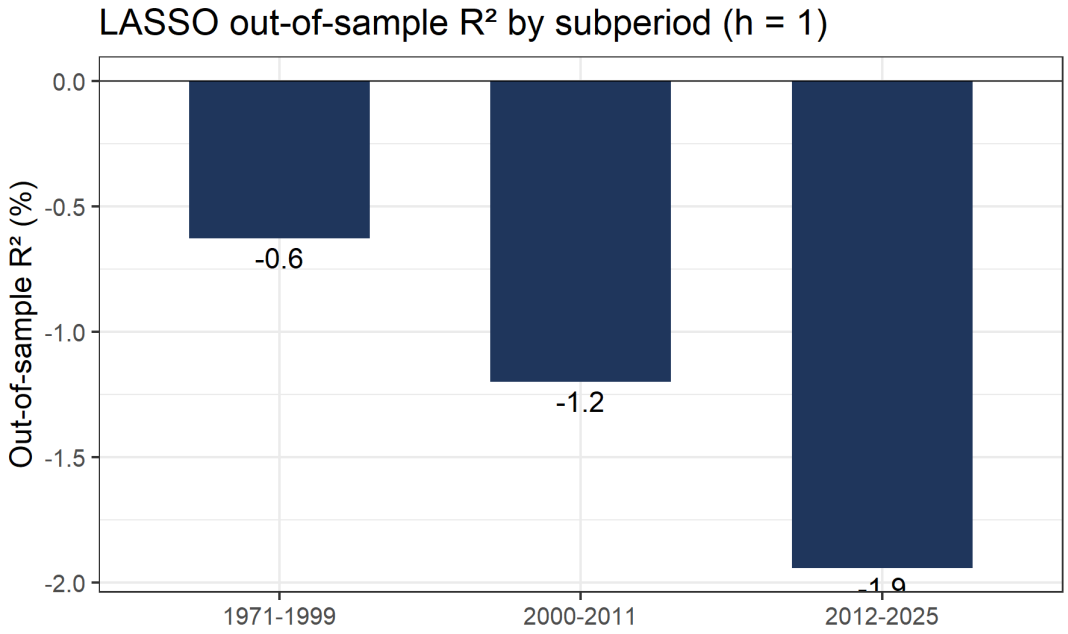


Figure 3: LASSO out-of-sample R^2 by subperiod at $h = 1$. Values are descriptive comparisons against the prevailing mean within each subperiod.

Table 5: Descriptive LASSO out-of-sample accuracy by subperiod ($h = 1$).

Subperiod	n	First target	Last target	R^2_{OOS} (%)
1971-1999	346	1971-03-01	1999-12-01	-0.63
2000-2011	144	2000-01-01	2011-12-01	-1.20
2012-2025	168	2012-01-01	2025-12-01	-1.94

R^2_{OOS} is computed against the prevailing mean within each subperiod. These comparisons are descriptive and not causal. Source files: `forecasts_h1.csv` and `panel.csv`. Generated by `R/13_figures.R`.

5 Conclusion

We apply a correctly time-aligned expanding-window bivariate design to the Neely et al. (2014) sample, with 178 paired observations available before the January 1966 forecast, and extend the

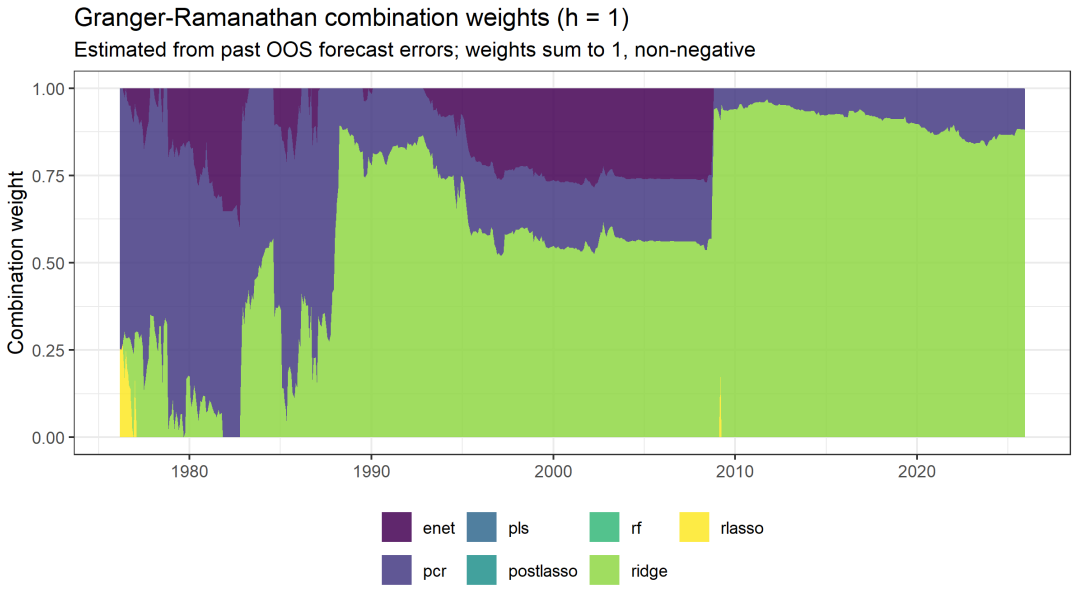


Figure 4: Granger–Ramanathan combination weights ($h = 1$), estimated from past out-of-sample forecast errors with $\sum_i w_i = 1$, $w_i \geq 0$.

analysis to 2025 with seven regularized machine learning models and two forecast combinations. Our computed technical signals match the published series in 99.98% of months. The three focal volume-rule estimates are positive but not significant at 5% in the original 1966–2011 evaluation period; none of the six volume rules is significant at 5%. On the extended sample, no model beats the prevailing historical mean at any horizon. Since the corrected analysis does not reproduce significance in the original period, these results do not establish whether previously reported predictability decayed after 2011. At $h = 1$ and 10 bps net of transaction costs, mean–variance ΔU is -0.39% p.a. for LASSO and -2.07% for post-LASSO. The economic strategies are evaluated separately from point-forecast accuracy; the complete fee sensitivity results are reported in Section 4.3 and Table 4.

Our study has clear limitations. It covers a single market (the US) with monthly data; the rolling window length and lag structure are fixed rather than selected; and the economic evaluation ignores short-sale constraints beyond the $[0, 1.5]$ weight bound and assumes the investor trades at the index level. Future work could extend the exercise to international equity markets, higher-frequency signals, or shrinkage schemes that combine the prevailing mean directly with the model forecast—a natural response to the persistent failure of point forecasts documented here.

A Predictor-set variants

Table 6 reports R^2_{OOS} at $h = 1$ for three predictor sets: the full design (84 macro-lag columns plus 14 technical signals), the 14 technical signals alone, and the 84 macro-lag columns alone. All nine reported R^2_{OOS} values are negative for each of the three predictor sets. These are descriptive comparisons; the bivariate regressions in Section 4.2 use one technical signal at a time and therefore estimate a different specification.

Table 6: Out-of-sample R^2 (%) at $h = 1$ by predictor set: full set (98 columns), technical signals only (14), and macro lags only (84).

Model	Full	Technical	Macro
rlasso	-1.06	-0.84	-1.03
postlasso	-2.83	-0.53	-2.80
ridge	0.12	-1.06	0.17
enet	-1.00	-1.13	-1.17
pcr	-1.29	-1.63	-2.15
pls	-1.24	-1.67	-1.25
rf	-15.53	-5.42	-15.13
equal_comb	-1.09	-1.10	-1.13
gr_comb	-0.54	-1.56	-0.49

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